

Turtle Games

Overview

Turtle Games, a leader in the global gaming industry, aims to boost sales and improve customer satisfaction. The company offers a diverse range of products, including books, board games, video games, and toys.

Business Objectives

Turtle Games aims to enhance overall sales performance by analysing customer engagement trends, loyalty point accumulation, segmentation opportunities and the strategic use of textual data in marketing.

Analytical Approach

The goal of this analysis is to use predictive modelling techniques and implement rigorous data cleaning and wrangling processes. In Python, single and multiple linear regression techniques are employed to predict customer loyalty. Linear regression analyses the relationship between loyalty points and demographic factors like age, education, income, and spending scores. Multiple linear regression assesses the combined effects of income and spending. Diagnostic tests for multicollinearity and heteroscedasticity, such as the Breusch-Pagan test, reveal heteroscedasticity in the data.

Decision tree analysis performed by encoding categorical variables, creating dummy variables, and converting loyalty labels into binary values. The pruned decision tree classifier demonstrates 99% accuracy and 97% precision on test data. Cluster

analysis using K-means, alongside elbow and silhouette methods, identifies five distinct customer segments based on income and spending.

Natural Language Processing (NLP) techniques analyse customer reviews to better inform marketing campaigns and operational strategies. The preprocessing of data includes text standardization, punctuation removal while preserving duplicates (50 duplicates in the review and 649 in the summary column), recognizing them as unique customer contributions, tokenization, and frequency distribution analysis. Sentiment analysis tools like TextBlob and Vader utilized, although positive comments such as 'Five Stars' classified as neutral.

In R, various libraries utilized for exploratory data analysis. This involves data wrangling, transforming the product column to a categorical data type, removing unnecessary columns, and customizing visualization themes. Visualizations include tables, histograms, and boxplots to examine data distributions, patterns, and outliers. Top-selling and underperforming products highlighted, suggesting areas for deeper sales analysis. Descriptive statistical analysis conducted to understand the dataset's characteristics. Skewness and kurtosis tests performed to assess the data's shape. Descriptive statistics indicate a nearly normal distribution with a slight tail. Visual examinations, including boxplot, histogram, and QQ-Plot, confirmed normality, supported by the Shapiro-Wilk test. Skewness and kurtosis tests revealed a slight right skewness and kurtosis below three (2.59), respectively, implying minimal outliers and successful process optimization. Further analysis involved correlation analysis and the creation of a multiple linear regression model. Additionally, a k-means clustering model implemented, although the clustering results appeared less distinct compared to those achieved in Python.

Visualisation and Insights

1. Loyalty Points Accumulation Analysis

The linear and multi-linear regression models explore the relationship between loyalty points and customer demographics. Insights from this analysis will help Turtle Games enhance customer engagement strategies and refine its loyalty programmes.

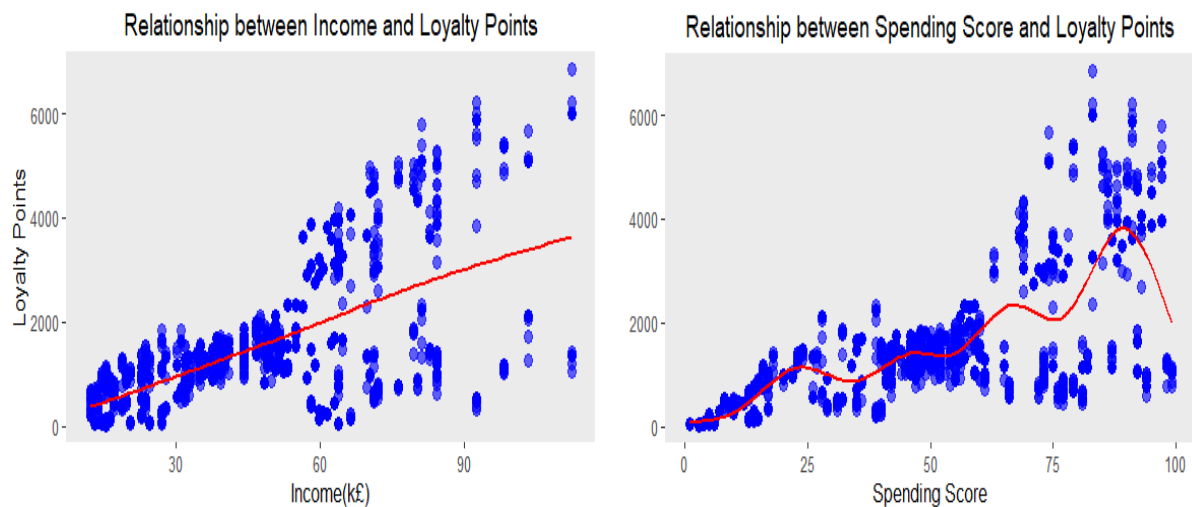


Figure 1. Income and loyalty account for 45% of variance, highlighting the importance of economic factors in loyalty schemes.

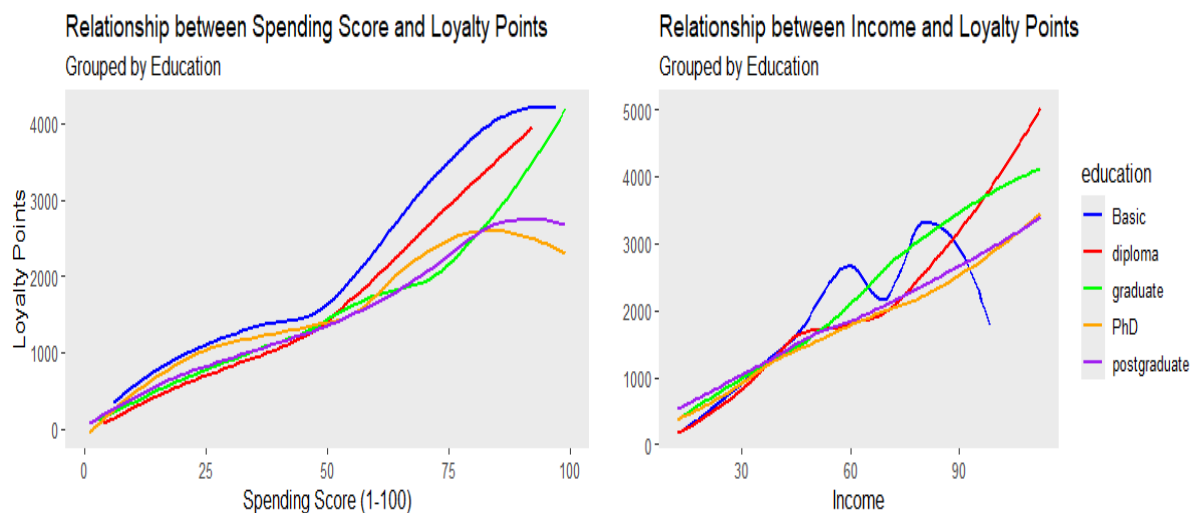


Figure 2. Customers with basic education demonstrate the highest loyalty compared to their spending score, despite having lower income.

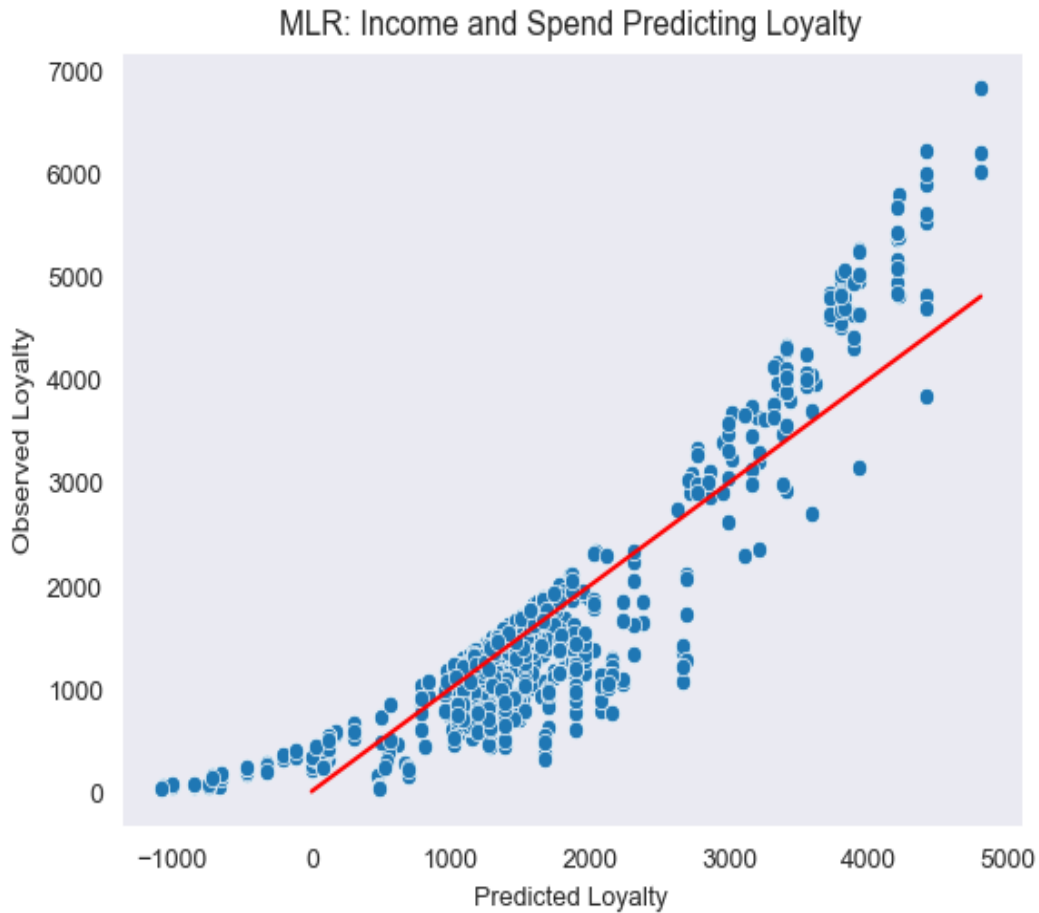


Figure 3. Multi-linear regression indicates that income and spending as significant predictors of loyalty, explaining 82.1% of the variability. Although no significant multicollinearity is observed, heteroscedasticity is present, indicating non-constant residual variability across observations.

Understanding loyalty's impact on sales is vital. Loyal customers drive revenue growth and spend more per transaction, emphasizing the importance of loyalty programmes. (PECB, 2024).

2. Exploring The Structure Using Decision Tree

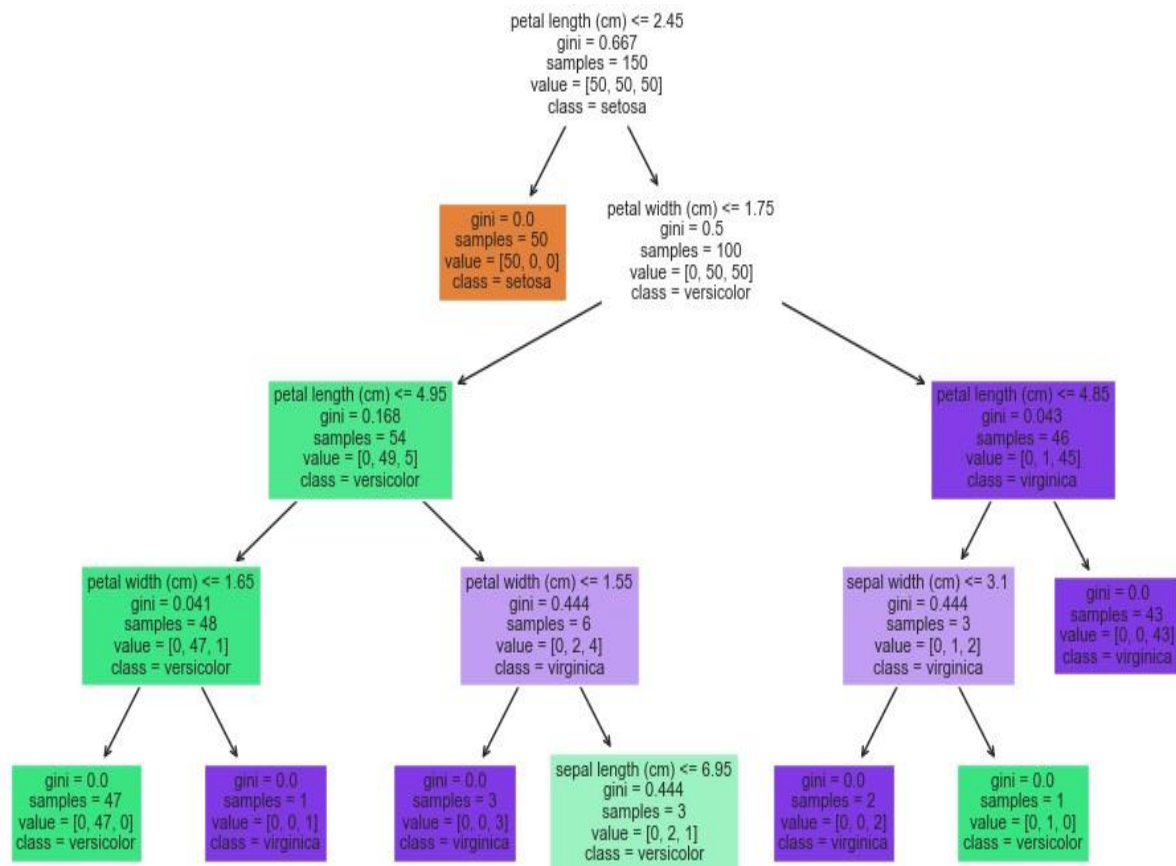


Figure 4. The decision tree model, with its high accuracy by 99% and precision by 97%, effectively predicts customer loyalty. This tool allows us to refine our understanding of how demographic factors influence loyalty accumulation.

Building a decision tree for a loyalty programme offers benefits: segmentation allows tailored offers based on customer attributes; targeted marketing increases engagement by understanding loyalty drivers; optimizing rewards aligns incentives with customer preferences. Decision trees aid in predicting churn, evaluating program effectiveness, and guiding proactive retention strategies. Leveraging decision trees in loyalty programs provides valuable insights into customer behaviour, improving the effectiveness of loyalty initiatives (Magee, J., n.d.).

3. Customer Segmentation and Targeting

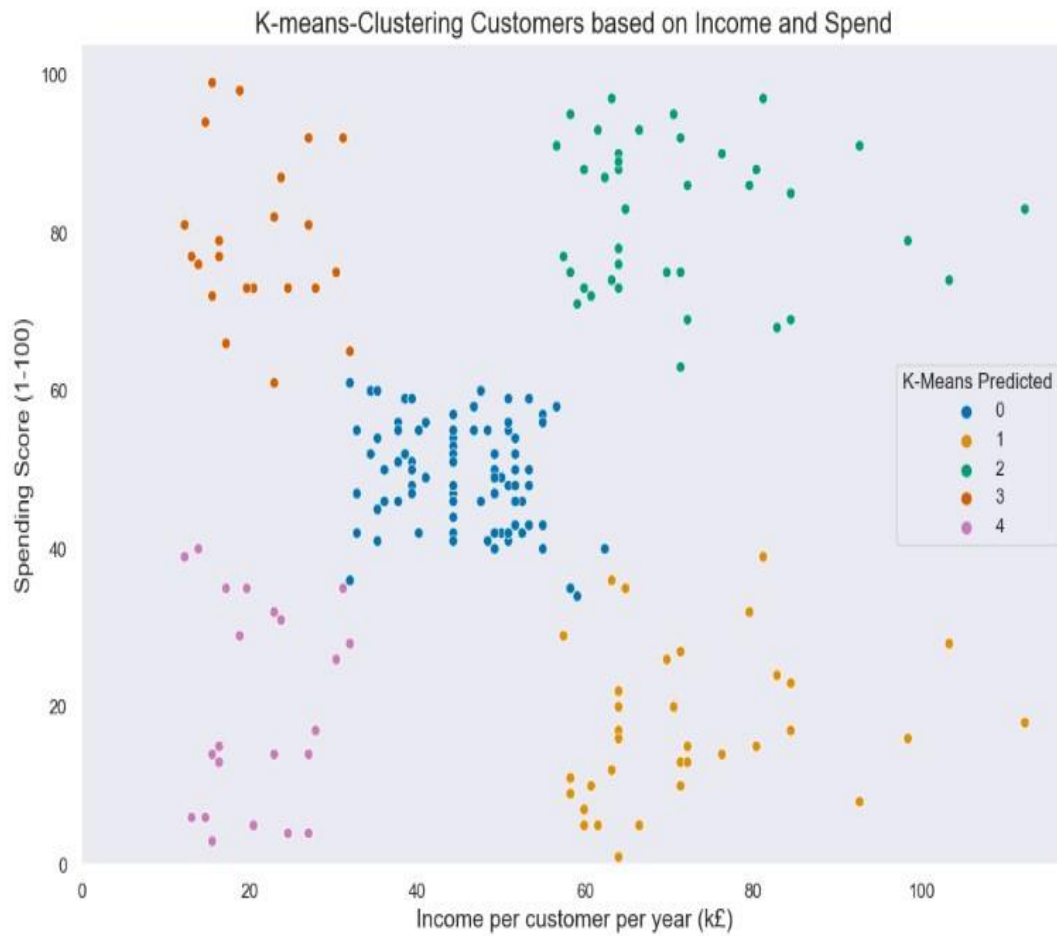


Figure 5. The application of the Elbow and Silhouette methods confirms that five clusters optimally represent the customer base. 0: moderate income and spending, 1: high income but low spending, 2: high income and high spending, 3: low income but high spending, 4: low income and spending. This segmentation facilitates targeted resource allocation and strategic marketing.

Understanding distinct clusters helps allocate resources effectively, enhance experiences, and seize market opportunities for sustainable growth. (Koirala, J. 2023).

4. Text Data Analysis for Marketing Insights

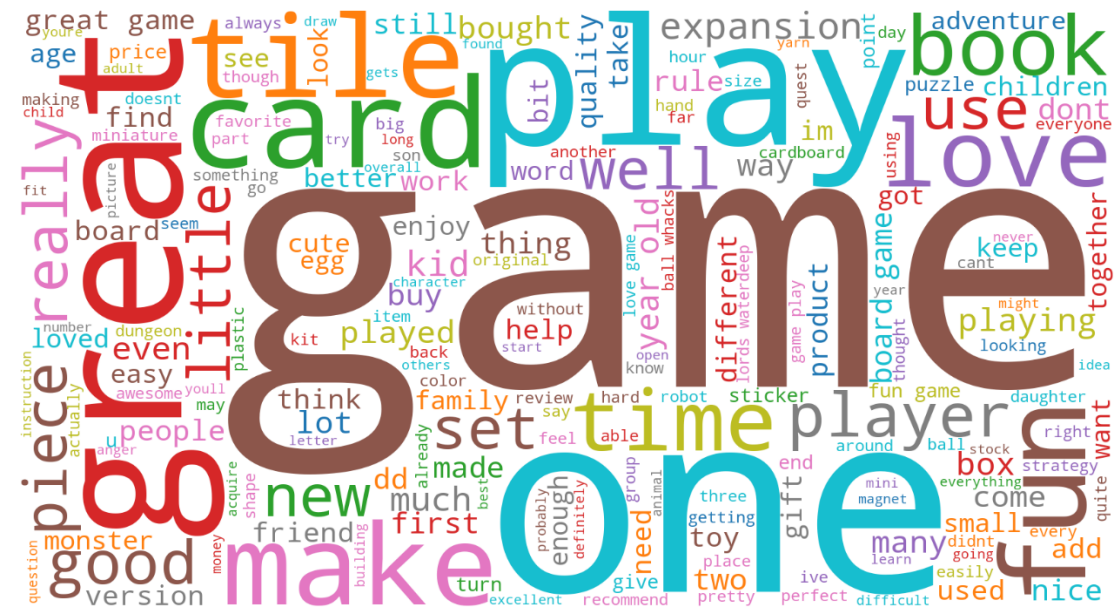


Figure 6. The consistent use of positive language in both the comprehensive reviews and the summaries indicates a strong favourability towards Turtle Game's products among customers.



Figure 7. Notably, in the summary reviews, the frequent pairing of 'five' and 'stars' underscores the prominent level of satisfaction expressed by customers.



Figure 8. The frequent mention of 'Quality' and 'Poor' alongside high satisfaction ratings indicates areas for potential improvement in product offerings (Otar, C. 2018).

Patterns and Predictions

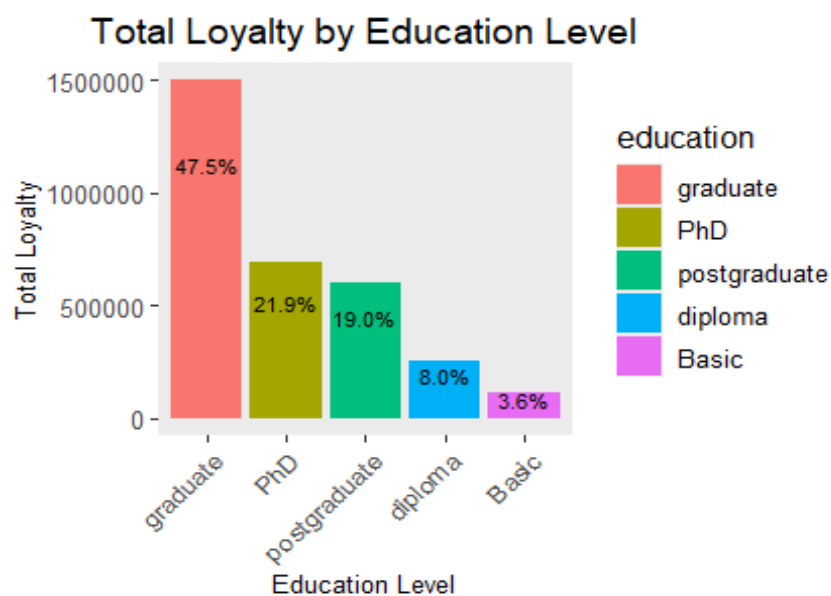


Figure 9. Educational Influence on Loyalty

- **Customer Loyalty and Engagement:** Predictive models indicate that income and education significantly influence loyalty point accumulation. By targeting high-income

and highly educated segments with tailored promotions and loyalty rewards, Turtle Games can increase customer retention and spending.

- **Segmentation for Targeted Marketing:** The clustering analysis has identified five distinct customer segments. Marketing strategies can be refined to address the unique needs and preferences of these groups, enhancing the effectiveness of promotional campaigns and potentially increasing market share.
- **Utilizing Customer Feedback:** NLP analysis of customer reviews highlights areas for product improvement and opportunities for positive reinforcement in marketing communications. By addressing commonly cited issues and leveraging praise, Turtle Games can improve product quality and customer satisfaction.
- **Data-Driven Product Development:** Descriptive statistics have confirmed the suitability of the loyalty points data for predictive modelling. This insight supports the development of new products and services that align with the preferences and behaviours of key customer segments, driving innovation and growth.

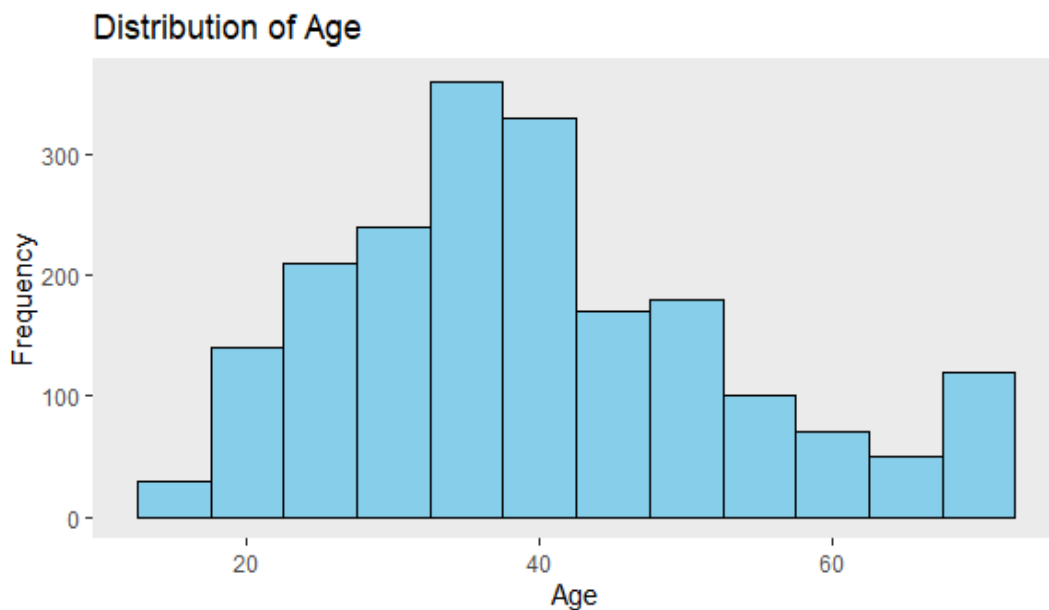


Figure 10. Age is not correlated with loyalty.

Conclusion

This comprehensive analysis provides Turtle Games with actionable insights into customer loyalty, segmentation, and behaviour patterns. By continuing to focus on data-driven strategies, Turtle Games can enhance customer experiences, optimize marketing efforts, and sustain growth.

References

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